

Assignment - 2

1. Discuss application firewalls & New Generation firewall w/ real world examples.

→ An application firewall filters traffic for specific apps (often Layer 7), blocking threats like SQL injected or cross-site scripting.

Example:

A bank protects its online portal w/ a web application firewall (WAF) such as Cloudflare or Akamai.

A Next-Generation Firewall (NGFW) is a security system that not only filters traffic based on IP & ports but also inspects application data, detects threats, and controls user activity in real-time.

Example: A company using an NGFW from Palo Alto Networks can block employees from using risky apps like torrents, detect malware in downloads & stop hackers trying to access internal systems all at once.

2. Analyze HTTP/HTTPS security issues like hidden manipulation and HSTS.

→ HTTP/HTTPS security issues :-

- Hidden Manipulation: In HTTP, data is unencrypted so attackers can intercept and modify it.

For example: fake login pages or public Wi-Fi.

- HTTPS issues: Safer, but can still be attacked via fake certificates or downgrade attacks (SSL stripping).

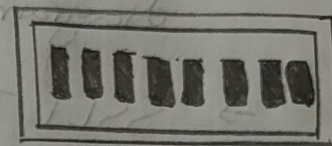
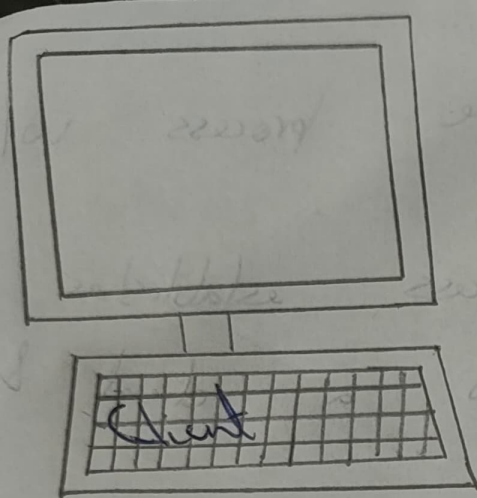
- HSTS: A security feature that forces browsers to use HTTPS only.

Sites like GOOGLE use it to prevent such attacks.

3. Explain SSL/TLS handshake process w/ diagram.

→ SSL/TLS handshake process establishes a secure, encrypted connectⁿ b/w a client & a server.

1. Client Hello: Browser sends supported TLS versions, cipher suites and a random number.
 2. Server Hello: Server selects settings, sends its digital certificate & its own random number.
 3. Certificate verificatⁿ: Client verifies the server's certificate (trust check).
 4. Key Exchange → Client creates a session key, encrypts it w/ the server's public key & sends it.
 5. Secure connectⁿ: Both sides use the session key to encrypt communicatⁿ.
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Server

Client Hello →

← Server Hello

← Certificate + Key Info

Key Exchange →

Secure Connect →